A two-pineapple envelope for split-core threshold random walks

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Abstract

Kemeny's constant is a stationary-target mean first-passage time for a finite random walk. A conjecture of Breen et al. proposes that, among threshold graphs of a fixed order, its maximum is attained by a pineapple graph. We prove an exact partial result for two infinite split-core families. For positive integers s, q, t and $n = s + q + t + 3$, each threshold code $01\,0^t1^s0^q1$ and $0\,1^s0^t1\,0^q1$ has Kemeny's constant strictly below the larger of the two pineapple values with core parameters $s + 1$ and $s + t + 1$. The proof reduces the general construction-code formula to four rational polynomials and uses exact positive-coefficient certificates. We also show that the tempting operation of filling the interior zero block is not monotone: it can increase, decrease, or leave Kemeny's constant unchanged. Thus the two-pineapple comparison is essential. The global pineapple conjecture and other split-core placements remain unresolved.

Keywords: Kemeny's constant; threshold graph; random walk; pineapple graph; exact symbolic computation; extremal graph theory

1 Introduction

Let a simple random walk move one edge per unit time on a finite connected graph. Kemeny's constant $\mathcal{K}$ is the expected time to reach a target drawn from the stationary distribution, and it does not depend on the initial vertex. It is simultaneously a transport statistic and a spectral quantity. The way in which graph structure controls $\mathcal{K}$ is subtle: adding edges changes degrees, transition probabilities, and stationary weights at the same time.

Threshold graphs provide a sharply structured setting in which this extremal question remains nontrivial. Breen et al. [1] derived an exact construction-code formula, studied several related walk parameters, and conjectured that a pineapple graph maximizes $\mathcal{K}$ among all threshold graphs of fixed order. Their openly available preprint reports computational verification through order 30, but the conjecture is universal.

This paper isolates two natural ways to split a dense core by a nonempty block of isolated-at-insertion vertices. Neither family can beat the pineapple envelope. More precisely, each member is strictly dominated by one of two pineapples of the same order. Which pineapple wins depends on the parameters; this dependence cannot be removed by a simple edge-filling monotonicity argument. The result excludes two three-parameter families from a search for a global counterexample, but it does not settle the full conjecture.

All calculations below are exact. The source formula was transcribed from arXiv:2310.08552v1, the version that was available for equation-level checking. The corresponding 2025 version of record was not available for a line-by-line comparison, so statements about proximity to prior work are intentionally bounded to the sources described in Section 5.

2 Threshold-walk model and the pineapple bound

A connected threshold graph on $n \geq 3$ vertices is encoded here by $c = (c_1, \ldots, c_n) \in \{0,1\}^n$ with $c_1 = 0$ and $c_n = 1$. A vertex coded by 1 is joined to every earlier vertex, while a vertex coded by 0 is joined to none of them. Equivalently,

$$A_{ii} = 0, \qquad A_{ij} = c_{\max(i,j)} \quad (i \neq j). \tag{1}$$

Let $d_i = \sum_j A_{ij}$, $D = \mathrm{diag}(d_i)$, and $m = \frac{1}{2}\sum_i d_i$. The non-lazy discrete-time walk has transition matrix $T = D^{-1}A$ and stationary distribution $\pi_i = d_i/(2m)$. If

$$H_{ij} = \mathbb{E}_i \inf\{k \geq 0 : X_k = j\}, \qquad H_{ii} = 0, \tag{2}$$

then

$$\mathcal{K}(G) = \sum_j \pi_j H_{ij} = \mathrm{Tr}\Big(I - T + \mathbf{1}\pi^{\mathsf{T}}\Big)^{-1} - 1 = \sum_{k=2}^{n} \frac{1}{1-\lambda_k(T)}. \tag{3}$$

The auxiliary first-hitting problem does not alter the original chain. In particular, periodic graphs are permitted and every step retains unit duration.

For $0 \leq r \leq n-2$, the pineapple graph $G_{n,r}$ has code $0\,1^r 0^{n-r-2} 1$. Its exact value is [1, Lemma 3.14]

$$F_n(r) = n - 4 + \frac{2}{r+2} + \frac{(n-1)(2r+3)}{2n + r^2 + r - 2}. \tag{4}$$

Write $B_n = \max_{0 \leq r \leq n-2} F_n(r)$, with the maximum taken over integer r. The global conjecture is $\mathcal{K}(G(c)) \leq B_n$ for every legal code c.

For completeness, the source reduction used in our calculation can be stated compactly. For $1 \leq i \leq n-1$, define vectors $w_i, z_i \in \mathbb{R}^n$ by

$$w_i = (0, 2, 4, \ldots, 2(i-1), -i(i-1), 0, \ldots, 0), \tag{5}$$

$$z_i = (0, \ldots, 0, i+1, 1, \ldots, 1), \tag{6}$$

where the distinguished entry is in coordinate $i+1$. Theorem 3.8 of Ref. [1] gives

$$\mathcal{K}(c) = n - 1 - \sum_{i=1}^{n-1} \frac{c_{i+1}}{z_i^{\mathsf{T}} c} + \sum_{i=1}^{n-1} \frac{(w_i^{\mathsf{T}} c)(2m - w_i^{\mathsf{T}} c)}{2m\, i(i+1) z_i^{\mathsf{T}} c}. \tag{7}$$

This expression and Eq. (4) were independently checked against literal full-graph hitting-time calculations before the new comparison was attempted.

3 Exact two-pineapple domination

For $s, q, t \in \mathbb{N}$ define the legal six-run codes

$$A_{s,q,t} = 01\,0^t 1^s 0^q 1, \qquad C_{s,q,t} = 0\,1^s 0^t 1\,0^q 1, \tag{8}$$

both of order $n = s + q + t + 3$. We use C for the second family to avoid confusing the graph with the scalar pineapple envelope B_n.

Theorem 1 (split-core envelope). For every $s, q, t \in \mathbb{N}$,

$$\mathcal{K}(A_{s,q,t}),\ \mathcal{K}(C_{s,q,t}) < \max\{F_n(s+1), F_n(s+t+1)\} \leq B_n. \tag{9}$$

3.1 Block reduction

On a constant-bit run $a \le i \le b$, both $z_i^{\mathsf{T}} c$ and $w_i^{\mathsf{T}} c$ are constant. If the run consists of ones, their values are respectively

$$b + 1 + \#\{\text{later ones}\}, \qquad 2P_a - a(a-1), \tag{10}$$

where P_a is the sum of the indices of earlier ones. On a zero run the corresponding values are the number of later ones and $2P_a$. The remaining factor telescopes:

$$\sum_{i=a}^{b} \frac{1}{i(i+1)} = \frac{1}{a} - \frac{1}{b+1}. \tag{11}$$

Thus Eq. (7) reduces each code in Eq. (8) to a rational function of s, q, t without a matrix diagonalization.

Set

$$\Delta_X^h = F_n(s+t+1) - \mathcal{K}(X), \qquad \Delta_X^\ell = F_n(s+1) - \mathcal{K}(X), \tag{12}$$

where $X = A_{s,q,t}$ or $C_{s,q,t}$. Exact simplification gives

$$\Delta_A^h = \frac{tH_A}{D_A^h}, \qquad \Delta_A^\ell = \frac{stL_A}{D_A^\ell}, \tag{13}$$

$$\Delta_C^h = \frac{tH_C}{D_C^h}, \qquad \Delta_C^\ell = \frac{tL_C}{D_C^\ell}. \tag{14}$$

Here H_X, L_X are integer polynomials fixed by these normalized expressions. Their complete expanded forms accompany this article as an exact machine-readable certificate. The denominators are

$$\begin{aligned}
D_A^h &= (s+1)(s+3)(s+t+3)(2q+s^2+2st+5s+2t+6) \\
&\quad \times (2q+s^2+2st+5s+t^2+5t+6), \\
D_A^\ell &= (s+1)(s+3)(s+t+3)(2q+s^2+5s+2t+6) \\
&\quad \times (2q+s^2+2st+5s+2t+6), \\
D_C^h &= (s+3)(s+t+3)(2q+s^2+5s+4t+6) \\
&\quad \times (2q+s^2+2st+5s+t^2+5t+6), \\
D_C^\ell &= (s+3)(s+t+3)(2q+s^2+5s+2t+6) \\
&\quad \times (2q+s^2+5s+4t+6).
\end{aligned}$$

Every displayed factor is positive on the stated domain.

3.2 Positive-coefficient certificates

For the first family, exact polynomial division yields

$$L_A + \alpha_A H_A = R_A, \qquad \alpha_A = \frac{s^2+4s+1}{s^2t+3s^2+4st+12s+3t+7} > 0. \tag{15}$$

The denominator of R_A is the denominator in α_A, and its numerator is an expanded polynomial with 49 nonzero monomials. Every coefficient is a strictly positive integer. The full polynomial

is retained verbatim in the supplementary certificate rather than abbreviated by a numerical fit. Consequently, if $H_A > 0$, Eq. (13) gives $\Delta_A^h > 0$; if $H_A \le 0$, Eq. (15) gives $L_A = R_A - \alpha_A H_A > 0$ and hence $\Delta_A^\ell > 0$.

For the second family, the large-s branch is already positive. With $s = u + 3$ and $u \ge 0$, direct expansion gives

$$\begin{aligned} L_C = {} & 2q^2u^2 + 14q^2u + 8q^2 + 6qtu^2 + 58qtu + 120qt \\ & + qu^4 + 22qu^3 + 177qu^2 + 616qu + 780q \\ & + 4t^2u^2 + 44t^2u + 112t^2 + 7tu^3 + 111tu^2 \\ & + 584tu + 1020t + 2u^4 + 45u^3 + 379u^2 + 1416u + 1980, \end{aligned} \tag{16}$$

which is strictly positive. The two remaining values of s have the exact certificates

$$L_C + \frac{3H_C}{4t+11} = \frac{2(71qt^2 + 403qt + 552q + 62t^3 + 514t^2 + 1356t + 1128)}{4t+11}, \qquad s = 1, \tag{17}$$

$$L_C + \frac{H_C}{5t+22} = \frac{349qt^2 + 3138qt + 7025q + 338t^3 + 3976t^2 + 14950t + 17600}{5t+22}, \qquad s = 2. \tag{18}$$

All coefficients and denominators in Eqs. (17)–(18) are positive. For $s \ge 3$, Eq. (16) directly gives $\Delta_C^\ell > 0$. For $s = 1, 2$, use the high-core gap if $H_C > 0$; otherwise Eqs. (17)–(18) imply $L_C > 0$, so the low-core gap is positive. This proves the strict first inequality in Eq. (9). Finally, $s + 1$ and $s + t + 1$ are legal integer pineapple parameters at the same n, proving the second inequality. □

The proof is computer-assisted only at the level of exact rational expansion and identity checking. It contains no floating-point sign decision, finite parameter cutoff, interpolation, or asymptotic extrapolation. The accompanying derivation script reconstructs Eqs. (13)–(18) from Eq. (7) and asserts the identities symbolically. A second implementation evaluates explicit code strings and selected literal full-graph fundamental matrices. Those finite checks protect against transcription and indexing errors; they are not substitutes for the all-parameter identities.

4 Why filling the core is not monotone

When $t = 1$, filling the interior zero turns either split-core graph into the pineapple with parameter $r = s + 2$. The sign of the resulting change is not fixed. For family A, it is the sign of

$$\begin{aligned} N_A(s,q) = {} & (4s^2 + 16s + 10)q^2 + (s^3 + 9s^2 + 18s + 4)q \\ & - (s^5 + 13s^4 + 66s^3 + 163s^2 + 193s + 84), \end{aligned} \tag{19}$$

because the exact filled-minus-defect difference is

$$\frac{4N_A(s,q)}{(s+1)(s+3)(s+4)(2q + s^2 + 7s + 8)(2q + s^2 + 7s + 12)}. \tag{20}$$

For family C, the corresponding sign polynomial is

$$\begin{aligned} N_C(s,q) = {} & (6s + 24)q^2 + (-s^3 - 5s^2 + 2s + 36)q \\ & - 2(s+3)(s+4)(s^2 + 6s + 6), \end{aligned} \tag{21}$$

and the difference is $(s + 1)N_C$ divided by

$$(s+3)(s+4)(2q + s^2 + 5s + 10)(2q + s^2 + 7s + 12). \tag{22}$$

Each sign polynomial is an upward-opening quadratic in q with negative constant term and hence has exactly one positive real root. The change is negative below that root and positive above it, with an integer equality possible at the root.

For example,

$$\mathcal{K}(011101) - \mathcal{K}(010101) = \frac{489}{110} - \frac{821}{180} = -\frac{229}{1980}, \tag{23}$$

whereas $s = 1, q = 4$ in the same family gives a positive change $11/840$. In the second family, $s = 12, q = 48$ gives exact equality, with both values $1055/16$. The alternative pineapple with $r = 13$ is nevertheless larger by $337/2640$, as Theorem 1 requires.

This reversal is the transport content behind the two-pineapple proof. Filling the core adds edges, but it also changes local escape probabilities and the degree-stationary target distribution. An argument that compares only the edge sets therefore misses the branch change in Eqs. (13) and (14).

5 Relation to prior work and limitations

The closest result is the construction-code formula and pineapple conjecture of Breen et al. [1]. Their paper supplies Eqs. (7) and (4), but the checked preprint does not state the two-family envelope proved here. Recent work on Braess cliques develops quotient and twin-vertex methods for changes in Kemeny's constant [2]; the checked results do not give the present split-core comparison. Effective-resistance matrices of weighted threshold graphs have also been obtained [3], but that work uses the combinatorial Laplacian and does not state this random-walk extremum. Accordingly, novelty is claimed only within the primary texts actually checked, not as an exhaustive priority certificate.

Theorem 1 has several sharp scope limits. It covers exactly the two placements in Eq. (8), with every run nonempty. It does not cover the other six-run arrangements, multiple separated zero blocks, or an arbitrary threshold code. It neither classifies equality cases of the global problem nor proves that every non-pineapple is strictly suboptimal. The full pineapple conjecture therefore remains unresolved. The model is an exact finite, unweighted, undirected, non-lazy walk; no continuum, thermodynamic, experimental, or perturbative claim is implied.

6 Conclusion

Two infinite split-core families obey the conjectured pineapple bound for all positive run lengths, and in fact each graph is strictly dominated by one of two explicitly identified pineapples of the same order. The need for a maximum of two comparators is structural: direct core filling has all three possible signs. A natural next step is to seek a finite pineapple envelope for the remaining six-run placements and then for two separated interior zero blocks. Such an extension requires new identities; the present theorem alone does not resolve the global extremal question.

Data and code availability

The exact derivation script, independent checker, machine-readable polynomial certificate, selected exact replay records, and reproduction instructions are included with the source package. No empirical dataset or stochastic sampling is used.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with problem development, symbolic code construction, exact calculations, computational checking, literature organization, and manuscript drafting. Its outputs remain subject to review, correction, and approval by the named human author before any submission. The named author must take responsibility for the content of any version that is submitted; this local draft does not represent that approval.

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